

ASSIGNMENT 4

1. What are local and global variables in the context of user-defined functions? Explain how these variables are scoped and utilized within functions, providing examples to illustrate their behavior.
2. Explain the concept of **recursion** in C programming. Provide examples of recursive functions and discuss their advantages and limitations compared to iterative approaches.
3. Describe the characteristics of **arrays, strings, and pointers** in C programming. Explain how arrays and strings are processed and manipulated, and discuss the role of pointers in accessing and managing memory.
4. Discuss the various storage classes in C, including **auto, extern, register, and static**. Explain their scopes, storage allocation, and lifetimes, providing examples to illustrate their usage in different scenarios.